

Figure 2: Peeling trace and counter-based delta on the running example.

(a) Peeling trace: 10 rounds × 6 CPI leaves (running example, $s = 3$)

round	pop	κ	L_1	L_2	L_3	L_4	L_5	L_6
1	v_8	1	C' Δ_h^p					
2	v_7	1	—					
3	v_{10}	1		C' Δ_h^p				
4	v_9	1		—				
5	v_2	3		—	C Δ_h^p			
6	v_1	3						
7	v_4	3			—	—	C	
8	v_5	3					C'	—
9	v_3	3			—	—	—	
10	v_6	3	—				—	—

(b) Delta zoom on Round 5: peel v_2 from L_3

Before:

$L_3 = (V_h = \{v_1\}, V_p = \{v_2, v_3, v_4\}), \eta = 2, p = 3$

encodes $\binom{3}{2} = 3$ triangles $\{v_1, v_2, v_3\}, \{v_1, v_2, v_4\}, \{v_1, v_3, v_4\}$

↓ peel v_2 (a pivot of L_3 , Case C)

After:

L_3 unchanged in memory; counter $p : 3 \rightarrow 2$, liveness still alive
effective encoding shrinks to $\binom{2}{2} = 1$ triangle $\{v_1, v_3, v_4\}$

Counter-based delta (Lemma 2):

$$\Delta_{\text{hold}} = \binom{p}{\eta} - \binom{p-d}{\eta} = \binom{3}{2} - \binom{2}{2} = 3 - 1 = 2$$

$$\Delta_{\text{pivot}} = \binom{p-1}{\eta-1} - \binom{p-d-1}{\eta-1} = \binom{2}{1} - \binom{1}{1} = 2 - 1 = 1$$

Support updates: η (hold) -2 ; v_3, v_4 (pivots) -1 each

No tree mutation, no hash op, $O(|V_h| + |V_p| - 1)$ work.

■ Case A: hold removed (dies)
 ■ Case C: pivot shrinks (alive)
 ■ Case C': pivot → death
 ■ already dead (skip)